RTL WITH VERILOGLAB4

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Q2) VGA Controller

Design Codes:

- **vga**

/home/cl4/cache/vga.sv (/tb_vga/DUT) - Default

```
module vga(
input clk,
output reg hsync,
output reg vsync,
output reg r,
output reg g,
output reg b
);
wire o_hsync;
wire o_vsync;
wire red, green, blue;
reg clk25MHz = 0;
reg [1:0] clk_div = 0;
always @(posedge clk) begin
clk_div <= clk_div + 1;
clk25MHz <= clk_div[1];
end
always@(*) begin
r = red;
g = green;
b = blue;
hsync = o_hsync;
vsync = o_vsync;
end
Simple_pattern top(
.clk25MHz(clk25MHz),
.o_hsync(o_hsync),
.o_vsync(o_vsync),
.o_red(red),
.o_green(green),
.o_blue(blue)
);
endmodule
```

- **Simple_pattern**

/home/cl4/cache/simple_pattern.v (/tb_vga/DUT/top) - Default

```
module Simple_pattern(
input wire clk25MHz,
output o_hsync,
output o_vsync,
output o_red,
output o_blue,
output o_green
);
reg [9:0] counter_x = 0;
reg [9:0] counter_y = 0;
wire [11:0] sram_addr;
wire [2:0] pixel_color;
reg [2:0] write_data;
reg [11:0] init_addr = 0;
reg init_write = 1;
always @(posedge clk25MHz) begin
if (init_write) begin
if (init_addr < 4095)
init_addr <= init_addr + 1;
else
init_write <= 0;
end
end
always @(*) begin
if (init_addr < 819)
write_data = 3'b100;
else if (init_addr < 1638)
write_data = 3'b001;
else if (init_addr < 2457)
write_data = 3'b101;
else if (init_addr < 3276)
write_data = 3'b010;
else
write_data = 3'b110;
end
always @(posedge clk25MHz) begin
```

```
if (counter_x < 799)
counter_x <= counter_x + 1;
else
counter_x <= 0;
end
always @(posedge clk25MHz) begin
if (counter_x == 799) begin
if (counter_y < 525)
counter_y <= counter_y + 1;
else
counter_y <= 0;
end
end
assign o_hsync = (counter_x < 10'd96) ? 1 : 0;
assign o_vsync = (counter_y < 10'd2) ? 1 : 0;
wire active_area = (counter_x > 144 && counter_x <= 783 && counter_y > 35 && counter_y <= 514);
wire [5:0] grid_x = (counter_x - 145) >> 3;
wire [5:0] grid_y = (counter_y - 36) >> 3;
assign sram_addr = init_write ? init_addr : {grid_y, grid_x};
sram_vga FRAME_BUFFER (
.clk(clk25MHz),
.reset(1'b0),
.chip_enable(1'b1),
.write_enable(init_write),
.read_enable(~init_write),
.address(sram_addr),
.data_in(write_data),
.data_out(pixel_color)
);
assign o_red = active_area ? pixel_color[2] : 1'b0;
assign o_green = active_area ? pixel_color[1] : 1'b0;
assign o_blue = active_area ? pixel_color[0] : 1'b0;
endmodule
```

- **clk_wiz_0**

/home/cl4/cache/clk_wiz_0_clk_wiz.v - Default

```
`timescale 1ps/1ps

module clk_wiz_0_clk_wiz
(
    // Clock outputs
    output        clk_out1,
    // Clock inputs
    input         clk_in1
);

    //----------------------------------------------------------------
    // Input buffering
    //----------------------------------------------------------------
    wire clk_in1_clk_wiz_0;

    IBUF clkin1_ibufg
    (
        .O(clk_in1_clk_wiz_0),
        .I(clk_in1)
    );

    //----------------------------------------------------------------
    // Clocking PRIMITIVE - PLLE2_ADV
    //----------------------------------------------------------------
    // Configuration:
    //   Input:  100 MHz
    //   VCO:    1000 MHz (100 MHz * 10)
    //   Output: 25 MHz  (1000 MHz / 40)
    //----------------------------------------------------------------

    // Internal signals
    wire        clk_out1_clk_wiz_0;
    wire        locked_int;
    wire        clkfbout_clk_wiz_0;
    wire        clkfbout_buf_clk_wiz_0;
```

- **PLL_8MHz**

/home/cl4/cache/PLL_25MHz.v - Default

```
//----------------------------------------------------------------------------
// None
//
//----------------------------------------------------------------------------
//  Output     Output      Phase    Duty Cycle   Pk-to-Pk     Phase
//   Clock     Freq (MHz)  (degrees)    (%)     Jitter (ps)  Error (p
//----------------------------------------------------------------------------
// clk_out1_____8.000______0.000______50.0______218.754_____98.575
//
//----------------------------------------------------------------------------
// Input Clock   Freq (MHz)    Input Jitter (UI)
//----------------------------------------------------------------------------
// __primary_________100.000____________0.010

`timescale 1ps/1ps

(* CORE_GENERATION_INFO = "clk_wiz_0,clk_wiz_v6_0_1_0_0,{component_na

module PLL_8MHz
 (
  // Clock out ports
  output        clk_out1,
 // Clock in ports
  input         clk_in1
 );

  clk_wiz_0_clk_wiz inst
  (
  // Clock out ports
  .clk_out1(clk_out1),
 // Clock in ports
  .clk_in1(clk_in1)
  );

endmodule
```

FPGA Commands:

```
cl4@CL4-29: ~/f4pga-examples/xc7
cl4@CL4-29:~$ cd ~/f4pga-examples
cl4@CL4-29:~/f4pga-examples$ export F4PGA_INSTALL_DIR=~/opt/f4pga
cl4@CL4-29:~/f4pga-examples$ export FPGA_FAM="xc7"
cl4@CL4-29:~/f4pga-examples$ source "$F4PGA_INSTALL_DIR/$FPGA_FAM/conda/etc/profile.d/conda.sh"
cl4@CL4-29:~/f4pga-examples$ conda activate xc7
(xc7) cl4@CL4-29:~/f4pga-examples$ export F4PGA_INSTALL_DIR=~/opt/f4pga
(xc7) cl4@CL4-29:~/f4pga-examples$ export FPGA_FAM="xc7"
(xc7) cl4@CL4-29:~/f4pga-examples$ cd xc7
(xc7) cl4@CL4-29:~/f4pga-examples/xc7$ make -C vga clean
make: Entering directory '/home/cl4/f4pga-examples/xc7/vga'
/home/cl4/f4pga-examples/xc7/vga/../../common/common.mk:39: *** Unsupported board type.  Stop.
make: Leaving directory '/home/cl4/f4pga-examples/xc7/vga'
(xc7) cl4@CL4-29:~/f4pga-examples/xc7$ TARGET="arty_100" make -C vga
make: Entering directory '/home/cl4/f4pga-examples/xc7/vga'
mkdir -p /home/cl4/f4pga-examples/xc7/vga/build/arty_100
cd /home/cl4/f4pga-examples/xc7/vga/build/arty_100 && symbiflow_synth -t vga  -v /home/cl4/f4pga-examples
4/f4pga-examples/xc7/vga/PLL_25MHz.v /home/cl4/f4pga-examples/xc7/vga/Simple_pattern.v -d artix7 -p xc7a1
[F4PGA] Running (deprecated) synth
-t
vga
-v
/home/cl4/f4pga-examples/xc7/vga/clk_wiz_0_clk_wiz.v
/home/cl4/f4pga-examples/xc7/vga/vga.v
/home/cl4/f4pga-examples/xc7/vga/PLL_25MHz.v
/home/cl4/f4pga-examples/xc7/vga/Simple_pattern.v
-d
artix7
-p
xc7a100tcsg324-1
-x
/home/cl4/f4pga-examples/xc7/vga/arty.xdc
```